

MA6.301 — Probability and Statistics
Assignment 3

Instructor: Naresh Manwani | Monsoon 2026 | Deadline: 27 septembet 11:59 PM

Total Marks: 100

Instructions

1. All questions are compulsory. Show all relevant working and clearly justify each step. State the axiom, definition, theorem, or result being used wherever applicable.
2. The assignment must be submitted in both hard copy and soft copy. The hard copy must be entirely handwritten; typed or printed submissions will not be accepted. The soft copy must be uploaded to Moodle as a single PDF named `rollnumber_a3.pdf`.
3. Group discussion and consultation of reference materials are encouraged for developing understanding and solving problems. However, the final solution and write-up must be your own individual work.
4. If you substantially use any textbook, lecture notes, website, paper, or other reference material, you must properly cite/reference the source in your submission.
5. Plagiarism or copying will not be tolerated. Any submission found to contain plagiarised or copied work may receive zero marks for the assignment/component and, in serious or repeated cases, an F grade in the course, as per course/institute policy.

Question 1. [5 marks]

Let X and Y be two jointly continuous random variables with the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{2}(3x + y), & 0 \leq x, y \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

and let the random vector $\mathbf{U} = \begin{bmatrix} X \\ Y \end{bmatrix}$.

- (a) Find the mean vector of $\mathbf{U}$, $\mathbb{E}[\mathbf{U}]$. [1]
- (b) Find the correlation matrix of $\mathbf{U}$, $R_{\mathbf{U}}$. [2]
- (c) Find the covariance matrix of $\mathbf{U}$, $C_{\mathbf{U}}$. [2]

Question 2. [5 marks]

Suppose that X , Y , and Z are three *independent* random variables. If $X, Y \sim \mathcal{N}(0, 1)$ and $Z \sim \text{Exponential}(1)$, find:

- (a) $\mathbb{E}[XY \mid Z = 1]$ [2]
- (b) $\mathbb{E}[X^2 Y^2 Z^2 \mid Z = 1]$ [3]

Question 3. [5 marks]

Let X and Y be jointly normal random variables with parameters $\mu_X = 0$, $\sigma_X^2 = 1$, $\mu_Y = -1$, $\sigma_Y^2 = 4$, and $\rho = -\frac{1}{2}$.

- (a) Find $P(X + Y > 0)$. [1]

- (b) Find the constant a such that $aX + Y$ and $X + 2Y$ are independent. [2]
- (c) Find $P(X + Y > 0 \mid 2X - Y = 0)$. [2]

Question 4. [5 marks]

Let X and Y be two independent $\mathcal{N}(0, 1)$ random variables. Define $Z = X + Y$ and $W = 2X - Y$. Find $\text{Cov}(Z, W)$.

Question 5. [5 marks]

Let X and Y be jointly normal random variables with parameters $\mu_X = 2$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 9$, and $\rho = -\frac{1}{2}$.

- (a) Find $\mathbb{E}[Y \mid X = 3]$. [1]
- (b) Find $\text{Var}(Y \mid X = 2)$. [2]
- (c) Find $P(X + 2Y \leq 5 \mid X + Y = 3)$. [2]

Question 6. [5 marks]

Let $X_1, X_2, \dots, X_n$ be independent random variables with $X_i \sim \text{Exponential}(\lambda)$. Define $Y = X_1 + X_2 + \dots + X_n$. As we will see later, $Y \sim \text{Gamma}(n, \lambda)$. Using this result or direct integration, show that

$$\mathbb{E}[Y] = \frac{n}{\lambda} \quad \text{and} \quad \text{Var}(Y) = \frac{n}{\lambda^2}.$$

Question 7. [5 marks]

Let X be a random variable with the following CDF:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < \frac{1}{4}, \\ x + \frac{1}{2}, & \frac{1}{4} \leq x < \frac{1}{2}, \\ 1, & x \geq \frac{1}{2}. \end{cases}$$

- (a) Find the generalized PDF of X , $f_X(x)$. [2]
- (b) Find $\mathbb{E}[X]$ using $f_X(x)$. [1]
- (c) Find $\text{Var}(X)$ using $f_X(x)$. [2]

Question 8. [5 marks]

Let X_1, X_2, X_3 be three i.i.d. Bernoulli(p) random variables and define $Y = X_1X_2 + X_2X_3$.

- (a) Find $\mathbb{E}[Y]$. [2]
- (b) Find $\text{Var}(Y)$. [3]

Question 9. [5 marks]

You are offered to play the following game. Roll a fair six-sided die once and observe X . You may stop and win X dollars, or roll again to observe Y , winning Y dollars instead. Let W denote your winnings.

- (a) What strategy maximises $\mathbb{E}[W]$? [3]
- (b) What is the maximum $\mathbb{E}[W]$ achievable under your strategy? [2]

Question 10. [5 marks]

Let X be a random variable with $\mathbb{E}[X] = \mu$. Define $f(\alpha) = \mathbb{E}[(X - \alpha)^2]$. Find the value of α that minimises f , and prove that it is indeed a minimum.

Question 11. [5 marks]

There are 1000 households in a town: 100 with one member, 200 with 2, 300 with 3, 200 with 4, 100 with 5, and 100 with 6 members, giving a total population of $N = 3300$.

- (a) A household is chosen uniformly at random. Let X be the number of members in that household. Find the PMF and $\mathbb{E}[X]$. [2]
- (b) A person is chosen uniformly at random. Let Y be the household size of that person's household. Find the PMF and $\mathbb{E}[Y]$. [3]

Question 12. [5 marks]

A continuous random variable has a Pareto(x_m, α) distribution if its PDF is

$$f_X(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, & x \geq x_m, \\ 0, & x < x_m, \end{cases}$$

where $x_m, \alpha > 0$. Let $X \sim \text{Pareto}(x_m, \alpha)$.

- (a) Find the CDF $F_X(x)$. [1]
- (b) Find $P(X > 3x_m \mid X > 2x_m)$. [2]
- (c) For $\alpha > 2$, find $\mathbb{E}[X]$ and $\text{Var}(X)$. [2]

Question 13. [5 marks]

Consider a gambler who at each round independently wins or loses the current bet with probabilities p and $1 - p$. When $p > \frac{1}{2}$, the *Kelly strategy* bets the fraction $2p - 1$ of the current fortune at each round. Starting with x units and employing the Kelly strategy, compute the expected fortune after n rounds.

Question 14. [5 marks]

(Two-Envelope Paradox.) Jill always chooses the option that maximises expected winnings. Harry secretly tosses a fair coin until tails appears; if there are n heads before the first tails, he places 10^n dollars in one envelope and 10^{n+1} dollars in the other. He offers Jill both envelopes; after opening one, she may switch for a \$1 fee.

- (a) If Jill's envelope contains \$100, what is the conditional probability the other contains \$1000? What is the probability it contains \$10? [2]
- (b) How much does Jill expect to win without switching? How much (net) if she does switch? [2]
- (c) Generalise the expected net winnings to the case where the first envelope contains 10^n dollars ($n \geq 0$). [1]

Question 15. [5 marks]

Let $X \sim \text{Uniform}(-2, 2)$ and define $Y = g(X)$, where

$$g(x) = \begin{cases} 1, & x > 1, \\ x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the generalized PDF and CDF of Y .

Question 16. [5 marks]

Let X and Y be independent Poisson random variables with rate parameters λ_1 and λ_2 , respectively. Find the conditional PMF of X given $X + Y = n$ (for non-negative integer n). Identify the name and parameters of this conditional distribution.

Question 17. [5 marks]

Let $X \sim \text{Uniform}(0, 1)$ and define $Y = -\frac{1}{\lambda} \ln(X)$ for some $\lambda > 0$. Derive the CDF and PDF of Y . What well-known distribution does Y follow?

Question 18. [5 marks]

Let $X, Y \sim \mathcal{N}(0, 1)$ be independent and define $U = X^2 + Y^2$. By first finding the CDF of U , derive its PDF. Identify the distribution of U .

Question 19. [5 marks]

A stick of length 1 is broken at a uniformly random point $X \sim \text{Uniform}(0, 1)$. Let Y denote the length of the longer piece. Express Y as a function of X and calculate $\mathbb{E}[Y]$.

Question 20. [5 marks]

Let X have the Laplace (double-exponential) PDF $f_X(x) = \frac{1}{2}e^{-|x|}$, $x \in (-\infty, \infty)$.

(a) Compute the moment generating function (MGF) $M_X(t)$. For which values of t is it valid?
[2]

(b) Using the MGF, compute $\text{Var}(X)$. [3]